

Tamoghna Ojha, *PhD*

CONTACT INFORMATION

Department of Mathematics and Computing
Indian Institute of Technology (ISM) Dhanbad
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RESEARCH INTEREST

AI/ML for Internet of Things, Future Networks, Non-terrestrial Networks, Wirelessly Powered Networks, Sensor-cloud, Resource Management.

WORK EXPERIENCE

SEPT 2024 – TILL DATE	Assistant Professor, IIT (ISM) Dhanbad, India <i>Department of Mathematics and Computing</i>
APR 2024 – SEPT 2024	Assistant Professor, BITS Pilani, Hyderabad, India <i>Department of Computer Science and Information Systems</i>
FEB 2023 – APR 2024	Assistant Professor, SRM University-AP, India <i>Department of Computer Science and Enginnering</i>

RESEARCH EXPERIENCE

SEPT 2022 – JAN 2023	ERCIM Post-Doctoral Research Fellow
OCT 2021 – JUL 2022	<i>Institute for Informatics and Telematics (IIT)</i> <i>National Research Council (CNR), Pisa 56124, Italy</i>
OCT 2022 – JAN 2023 (3 MONTHS)	Research Consultant <i>Department of Information Engineering</i> <i>University of Pisa, Pisa 56122, Italy</i>
JAN 2021 – SEPT 2021	Post-Doctoral Research Fellow <i>Institute for Informatics and Telematics (IIT)</i> <i>National Research Council (CNR), Pisa 56124, Italy</i>
NOV 2013 – JUL 2017	Senior Research Fellow <i>Sponsored Research and Industrial Consultancy</i> <i>Indian Institute of Technology Kharagpur, West Bengal 721302, India</i>
NOV 2010 – MAR 2013	Junior Project Assistant <i>Sponsored Research and Industrial Consultancy</i> <i>Indian Institute of Technology Kharagpur, West Bengal 721302, India</i>

TEACHING EXPERIENCE

IIT (ISM) DHANBAD SEPT 2024 – TILL DATE	Computer Networks (Theory & Lab); for UG, Autumn 2026 Operating Systems (Theory & Lab); for PG, Spring 2026 Computing Lab; for UG, Autumn 2025 Database Management Systems (Lab); for UG/PG, Spring 2025 Advanced DBMS (Lab); for PG, Spring 2025 Software Engineering (Lab); for UG, Spring 2025
BITS PILANI, HYDERABAD (APR 2024 – SEPT 2024)	Advanced Operating Systems (Theory and Lab); for PG, Autumn 2024 Operating Systems (Theory); for UG, Summer 2024

SRM UNIVERSITY-AP (FEB 2023 – APR 2024)	Data Structures (Theory and Lab); for UG, Spring 2024 IoT Design Protocols (Theory); for UG, Spring 2024 Computer Networks (Theory and Lab); for UG, Autumn 2023 Service Oriented Computing (Theory and Lab); for UG, Autumn 2023 Data Structures (Theory and Lab); for UG, Spring 2023
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SPONSORED RESEARCH PROJECTS

- **Title:** Resilient Edge-LoRa System for Early Gas Hazard Detection in Coal Mines and Adjacent Communities, **Sponsoring Agency:** Jharkhand Council on Science, Technology and Innovation (JCSTI), **Amount:** INR 5.955 Lakhs, **Duration:** 2026-29, **Role:** PI.
- **Title:** Revolutionizing Autonomous Navigation through Occlusion-Resilient Autonomous Driving Using Intelligent Perception Systems, **Sponsoring Agency:** IIT (ISM) Dhanbad (Collaborative Grant), **Amount:** INR 60.00 Lakhs, **Duration:** 2025-27, **Role:** Co-PI.
- **Title:** Improving The Performance of IEEE 802.11ah UAV Networks In Mission-Critical Defense Environments, **Sponsoring Agency:** NMICPS TiHAN-IIT Hyderabad, **Amount:** INR 21.91 Lakhs, **Duration:** 2025-26, **Role:** Co-PI.
- **Title:** Energy-Efficient Federated Deep Reinforcement Learning for Multi-UAV Edge Intelligence Using Tiny Models, **Sponsoring Agency:** IIT (ISM) Dhanbad (FRS Grant), **Amount:** INR 19.998 Lakhs, **Duration:** 2025-27, **Role:** PI.
- **Title:** AI-Powered Vision Systems for Low-light and Low-visibility Underground Mining Environments, **Sponsoring Agency:** TEXMiN Foundation (DST TiH), **Amount:** INR 17.35 Lakhs, **Duration:** March 2025 - September 2026 (18 months), **Role:** Co-PI.
- **Title:** Special Lab Setup Grant: Computing for Secure and Intelligent Networks (COSINE), **Sponsoring Agency:** IIT (ISM) Dhanbad (Special grant), **Amount:** INR 29.97 Lakhs, **Role:** PI.
- **Title:** Using Edge Intelligence for Resource Allocation in Wirelessly Powered UAV-IoT Network, **Sponsoring Agency:** SRM University-AP (Seed grant), **Amount:** INR 15.24 Lakhs, **Duration:** Oct 2023 - Sept 2025 (24 months), **Role:** PI.
- **Title:** Breaking the Barriers of Skin Disease Diagnosis with Computational Imaging and Artificial Intelligence, **Sponsoring Agency:** SINE, IIT Bombay and Intel Inc. (Plugin 2 Startup Cohort), **Amount:** INR 10.00 Lakhs, **Duration:** Oct 2018 - Jun 2019 (9 months), **Role:** Co-PI.
- **Title:** Multispectral Optical Imaging and Computing Technologies for Realtime in-situ Functional Characterization and Monitoring of Cutaneous Wound Healing Progression, **Sponsoring Agency:** BIRAC, DBT, Govt. of India (BIG grant), **Amount:** INR 41.79 Lakhs, **Duration:** Aug 2015 - Jan 2017 (18 months), **Role:** Co-PI.

EDUCATION

JUL 2014 – FEB 2021	Doctor of Philosophy in COMPUTER SCIENCE AND ENGINEERING Indian Institute of Technology Kharagpur, India Thesis Title: <i>Provisioning Sensors-as-a-Service in Sensor-cloud-based Internet of Things</i> Advisors: Prof. Sudip Misra (FIEEE, FACM, FNAE, FNASc) and Prof. Narendra Singh Raghuwanshi (FNAE, FNAAS) CGPA: 8.05/10 Thesis defended: 13-Nov-2020
DEC 2010 – JUL 2014	Master of Science by Research in INFORMATION TECHNOLOGY Indian Institute of Technology Kharagpur, India Thesis Title: <i>Architecture and Localization for Underwater Sensor Networks</i> Advisor: Prof. Sudip Misra (FIEEE, FACM, FNAE, FNASc) CGPA: 9.6/10
SEPT 2008 – FEB 2009	Post Graduate Diploma in EMBEDDED SYSTEMS DESIGN Center for Development of Advanced Computing, Mohali, India Project Topic: <i>A Real-time application for peripheral controlling and monitoring</i> Advisor: Ms. Sonia Dosanjh Marks: 80.50%
AUG 2004 – AUG 2008	Bachelor of Technology in ELECTRONICS AND COMMUNICATION ENGINEERING West Bengal University of Technology, Kolkata, India Institute: Haldia Institute of Technology, Haldia

Thesis Title: *Microcontroller based Infra-Red Tracking Robot*

Advisor: Dr. Mousiki Kar

DGPA: 7.89/10

JUL 2004 Higher Secondary (Board: WBCHSE)

Institute: Vidyasagar Vidyapith, Midnapore, West Bengal, India

Subjects: Physics, Chemistry, Mathematics, Biology, Bengali, English

Marks: 71.60%

MAY 2002 Secondary (Board: WBBSE)

Institute: Pirakata High School, Paschim Medinipur, West Bengal, India

Subjects: Bengali, English, Mathematics, Physical Science, Life Science, History, Geography

Marks: 83.38%

AWARDS AND SCHOLARSHIPS

- Dec 2024: **Best Paper Award** in IEEE Proceeding on 22nd OITS International Conference on Information Technology.
- Jun 2023: Elevated to **IEEE Senior Member** grade.
- Oct 2022 - Jan 2023: *Research Consultancy* at the Dept. of Information Engineering, University of Pisa, Italy.
- Oct 2021 - Jul 2022, Sept 2022 - Jan 2023: “**Alain Bensoussan**” **Post-Doctoral Research Fellowship** from European Research Consortium for Informatics and Mathematics (ERCIM).
- Jan 2021 - Sept 2021: **Post-Doctoral Research Fellowship** from National Research Council (CNR) Italy.
- Aug 2019 - May 2020: **Research Assistantship** from Indian Institute of Technology Kharagpur.
- Aug 2017 - Jul 2019: **Senior Research Fellowship** from MHRD, Govt. of India.
- Dec 2016: **Richard E Merwin Student Scholarship** from IEEE Computer Society. (Award: USD 1,000)
- Nov 2013 - Jul 2017: **Senior Research Fellowship** from ITRA, Govt. of India.
- Dec 2013: **Winner of GE Edison Challenge 2013**, GE John F. Welch Technology Center, Bangalore. (Award **INR 10.00 Lakhs**)
- Oct 2002: “Ardhendu Sekhar Sarkar Memorial Prize” for securing **1st position at Block** (Salboni, Paschim Medinipur) level in Secondary (Madhyamik) exam 2002.
- 2000–2004: **Scholarship** from Foundation for Excellence Inc., USA.

TRAVEL GRANTS

- December 2018: **Best Conference Travel Grant** support from IIT Kharagpur for presenting my paper at IEEE GLOBECOM 2018 in Abu Dhabi, UAE.
- December 2014: **International Travel Grant** from DST (SERB), Govt. of India for attending IEEE CloudCom, Singapore.
- December 2014: **Conference Travel Grant** from IEEE CloudCom 2014. (Award: SGD 1,000)

OTHER AWARDS

- October 2017: *IEEE TechSym 2016* (I served as Organizing Chair) was selected for **2017 Darrel Chong student activity award** in *GOLD* category.

STUDENT GUIDANCE

PHD STUDENT	Sanjeev Kumar (July 2026–)	Topic: AI-native Networking
	Arjab De (July 2025–)	Topic: Futuristic Intelligent Networks
M.TECH/M.SC.	Pintu Shaw (2025-27)	Topic: Agentic AI for Intelligent Networks
	Debanshu Jana (2025-27)	Topic: Digital Twin for Mining
	Subhadeep Nandi (2025-27)	Topic: UAV-based EdgeAI for Mining
	Kimsie Phan (2024-26)	Topic: Federated Learning for Resource-constrained IoT
	Scaria P. Anil (2024-26)	Topic: Semantic Communication and Resource Allocation

UG/INTERN (Selected)	Tarun Kr. Das (2024-26)	Topic: Non-terrestrial Networks
	Shrish Jadhav (2024-26)	Topic: Blockchain for IoT
	Anjali Ojha	Topic: Resource-constrained EdgeAI
	Mannan Golchha (2026)	Topic: Agentic AI
	Gargi Kedia	Topic: Multimodal EdgeAI
	Manjil Nepal	Topic: Federated Learning for Resource-constrained IoT

PATENTS

- D. Sheet, K. Basak, **T. Ojha**, S. P. K. Karri, "Multispectral Optical Imaging Device and Computational Techniques for Contactless Functional Imaging of Skin", *Indian Patent Published*, Application No. 201731022695, Published: 4 January 2019, Applicant: SkinCurate Research Private Limited, Filed: 28 June 2017.
- S. Misra, A. Roy, P. Kar, S. Goswami, **T. Ojha**, "An Adverse Environmental Effect Resistant Seamless Wireless Sensor Network System", *Indian Patent No. 480946*, **Granted: 12 December 2023**.

BOOKS / CHAPTERS

- **T. Ojha**, M. M. Hussain, S. Bera, N. Ahmed, S. Misra (eds.), "Edge-enabled 6G Networking - Foundations, Technologies, and Applications", Springer Nature, May 2026. (ISBN: 978-3032190413)
- A. Ghosh and **T. Ojha**, "mm-Wave Technology for 6G Communication", In "Edge-enabled 6G Networking - Foundations, Technologies, and Applications", **T. Ojha**, M. M. Hussain, S. Bera, N. Ahmed, S. Misra (eds.), Springer Nature, May 2026.
- N. Ahmed, **T. Ojha**, M. M. Hussain, S. Bera, "Edge-Enabled 6G Networking: Transforming IoT for a Hyperconnected Future", **T. Ojha**, M. M. Hussain, S. Bera, N. Ahmed, S. Misra (eds.), Springer Nature, May 2026.
- N. Choudhury, J. Dave, **T. Ojha**, "LoRaWAN Based IoT to Satellite Communication for Integrated Terrestrial and Non-terrestrial Networks", In "Edge-enabled 6G Networking - Foundations, Technologies, and Applications", **T. Ojha**, M. M. Hussain, S. Bera, N. Ahmed, S. Misra (eds.), Springer Nature, May 2026.

JOURNAL PUBLICATIONS

1. C. Amala, B. Subbarao, M. Sivaji, **T. Ojha**, B. B. Das, S. P. Mohanty, S. K. Ram, "Everything You Wanted to Know about Secure Energy Harvesting Techniques for IoT Devices-A Review", *IEEE Sensors Journal*, vol. 25, no. 22, pp. 41027 - 41039, 2025. (SCI: Q1, I.F.: 4.5)
2. **T. Ojha**, T.P. Raptis, A. Passarella, M. Conti, "Wireless Power Transfer with Unmanned Aerial Vehicles: State of the Art and Open Challenges", *Pervasive and Mobile Computing (Elsevier)*, vol. 93, pp. 101820, 2023. (SCI: Q1, I.F.: 3.848)
3. **T. Ojha**, T. P. Raptis, M. Conti, A. Passarella, "Balanced Wireless Crowd Charging with Mobility Prediction and Social Awareness", *Computer Networks (Elsevier)*, vol. 211, pages 108989, 2022. (Invited Submission) (SCI: Q1, I.F.: 5.493)
4. S. Misra, M. Tiwari, **T. Ojha**, Y. Raj, "PANDA: Preference-based Bandwidth Allocation in Fog-enabled Internet of Underground-Mine Things", *IEEE Systems Journal*, vol. 15, no. 4, pp. 5144 - 5151, 2021. (SCI: Q1, I.F.: 4.802)
5. **T. Ojha**, S. Misra, N. S. Raghuwanshi, "Internet of Things for Agricultural Applications: The State of the Art", *IEEE Internet of Things Journal*, vol. 8, no. 14, pp. 10973 - 10997, 2021. (SCI: Q1, I.F.: 10.238)
6. S. Misra, **T. Ojha**, P. Madhusoodhanan, "SecRET: Secure Range-Based Localization with Evidence Theory for Underwater Sensor Networks", *ACM Transactions on Autonomous and Adaptive Systems*, vol. 15, no. 1, pp. 1 - 26, 2021. (SCI: Q2, I.F.: 1.913)
7. **T. Ojha**, S. Misra, M. S. Obaidat, "SEAL: Self-adaptive AUV-based Localization for Sparsely Deployed Underwater Sensor Networks", *Computer Communications (Elsevier)*, vol. 154, pp. 204 - 215, 2020. (SCI: Q1, I.F.: 5.047)
8. **T. Ojha**, S. Misra, N. S. Raghuwanshi, H. Poddar, "DVSP: Dynamic Virtual Sensor Provisioning in Sensor-Cloud based Internet of Things", *IEEE Internet of Things Journal*, vol. 6, no. 3, pp. 5265 - 5272, 2019. (SCI: Q1, I.F.: 10.238)
9. **T. Ojha**, S. Misra, N. S. Raghuwanshi, "Sensing-cloud: Leveraging the Benefits for Agricultural Applications", *Computers and Electronics in Agriculture*, vol. 135, pp. 97 - 106, 2017. (SCI: Q1, I.F.: 6.757)

10. A. K. Mandal, S. Misra, **T. Ojha**, M. K. Dash, and M. S. Obaidat, "Oceanic Forces and their Impact on the Performance of Mobile Underwater Acoustic Sensor Networks," *International Journal of Communication Systems (Wiley)*, vol. 30, no. 1, pp. e2882, 2017. (SCI: Q2, I.F.: 1.882)
11. S. Misra, S. Bera, **T. Ojha**, H. Mouftah, A. Anpalagan, "ENTRUST: Energy Trading Under Uncertainty in Smart Grid Systems", *Computer Networks (Elsevier)*, vol. 110, pp. 232 - 242, 2016. (SCI: Q1, I.F.: 5.493)
12. A. K. Mandal, S. Misra, M. K. Dash, and **T. Ojha**, "Performance Analysis of Distributed Underwater Wireless Acoustic Sensor Networks in the Presence of Internal Solitons," *International Journal of Communication Systems (Wiley)*, vol. 29, no. 13, pp 1940 - 1955, 2016. (SCI: Q2, I.F.: 1.882)
13. **T. Ojha**, S. Misra, N. S. Raghuwanshi, "Wireless Sensor Networks for Agriculture: The State-of-the-Art in Practice and Future Challenges", *Computers and Electronics in Agriculture*, vol. 118, pp. 66 - 84, 2015. (**Most cited and most downloaded** paper of this journal during 2016–19.) (SCI: Q1, I.F.: 6.757)
14. A. K. Mandal, S. Misra, **T. Ojha**, M. K. Dash, and M. S. Obaidat, "Effects of Wind-induced Near-surface Bubble Plumes on the Performance of Underwater Wireless Acoustic Sensor Networks", *IEEE Sensors Journal*, vol. 16, no. 11, pp. 4092 - 4099, 2015. (SCI: Q1, I.F.: 4.325)
15. S. Misra, S. Bera, **T. Ojha**, L. Zhou, "ENTICE: Agent-Based Energy Trading with Incomplete Information in the Smart Grid", *Journal of Network and Computer Applications*, vol. 55, pp. 202 - 212, 2015. (SCI: Q1, I.F.: 7.574)
16. S. Misra, **T. Ojha**, and A. Mondal, "Game-theoretic Topology Control for Opportunistic Localization in Sparse Underwater Sensor Networks," *IEEE Transactions on Mobile Computing*, vol. 14, no. 5, pp. 990 - 1003, 2015. (SCI: Q1, I.F.: 6.075)
17. S. Misra, S. Bera, and **T. Ojha**, "D2P: Distributed Dynamic Pricing Policy in Smart Grid for PHEVs Management," *IEEE Transactions on Parallel and Distributed Systems*, vol. 26, no. 3, pp. 702 - 712, 2014. (SCI: Q1, I.F.: 3.757)
18. **T. Ojha**, M. Khatua, and S. Misra, "Tic-Tac-Toe-Arch: A Self-organizing Virtual Architecture for Underwater Sensor Networks," *IET Wireless Sensor Systems*, vol. 3, no. 4, pp. 307 - 316, 2013. (SCI: Q2, I.F.: 2.51)

CONFERENCE PUBLICATIONS

1. Afa, A. Kalita, **T. Ojha**, "Traffic Prediction and Load-Aware Adaptive RAW Slot Allocation for IEEE 802.11ah IoT Networks", *TENCON 2026 - 2026 IEEE Region 10 Conference*, Bali, Indonesia, 2026.
2. A. Gupta, S. Kumawat, M. Verma, **T. Ojha**, A. Galav, "FADER: Frequency ADaptive implicit nEuRal representations", *11th National Conference on Computer Vision, Pattern Recognition, Image Processing and Graphics (NCVPRIPG)*, July 2026, Jaipur, India.
3. T. P. Raptis and **T. Ojha**, "Gait-Diverse Avatar Selection for Realistic Synthetic Crowds in the Metaverse", in *IEEE Computers, Software, and Applications Conference (COMPSAC)*, Madrid, Spain, July 2026.
4. T. K. Das and **T. Ojha**, S. Bhosale, S. Karmakar, S. Halder, S. Barik, "A Quantum Reinforcement Learning Framework for Sustainable Internet of Vehicles toward Future Electric Mobility", in *IEEE International Conference on Sustainable Energy and Future Electric Transportation (SEFET)*, VNIT Nagpur, July 2026.
5. T. K. Das, **T. Ojha**, N. Choudhury, A. Kaushik, "A Hybrid FL-MARL Framework for Optimal Trajectory Control and Resource Allocation in Integrated Terrestrial and Non-Terrestrial Networks", in *IEEE ICC Workshops*, Glasgow, Scotland, May 2026.
6. A. Gorrela, S. V. Gudipudi, N. Choudhury, A. Hazarika, **T. Ojha**, A. Jindal, "Frame-Aware Resilient ADR for Relay-Connected LoRaWAN IoT Networks", in *IEEE ICC Workshops*, Glasgow, Scotland, May 2026.
7. T. Das, **T. Ojha**, N. Choudhury, A. Kaushik and K. Gupta, "QoS-aware Joint Optimization of Task Offloading and Resource Allocation in Integrated Terrestrial Non-Terrestrial Network", in *Proceedings of IEEE ANTS*, December 2025.
8. S. Anand, A. Gorrela, R. Rahman, N. Choudhury, A. Hazarika, D. Choudhury, **T. Ojha**, "Delay-Bounded Adaptive MAC for IEEE 802.15.4e DSME Networks: Enhancing Resilience under Bursty and Dynamic IoT Traffic", in *Proceedings of IEEE ANTS*, December 2025.
9. M. Vaiyata, **T. Ojha**, N. Choudhury, J. Dave and P. Lorenz, "Distributed RL-Based UAV Selection for Reliable Task Execution in Aerial IoT Edge Networks", in *Proceedings of IEEE GLOBECOM Workshops*, December 2025.
10. R. Rahman, N. Choudhury, W. Xiang and **T. Ojha**, "FlexiFrame: Adaptive Slotframe Scheduling for 6TiSCH-Based Smart City IoT Networks", in *Proceedings of IEEE GLOBECOM Workshops*, December 2025.

11. N. Roshni, S. Prabhugari, R. Soni, A. S. Manish Reddy, A. Soni, J. Dave and **T. Ojha**, "SDN Security Enhancement through Machine Learning based Packet Detection and Rate-Limiting", in *IEEE INDISCON*, Rourkela, India, August 2025.
12. S. Ravikanti, J. Dave, H. Dong, I. Gondal, N. Choudhury, **T. Ojha** and T. P. Raptis, "Avatar-Centric Gait Authentication Framework for Secure Metaverse", in *15th Workshop on Management of Cloud and Smart City Systems (with IEEE ISCC 2025)*, Bologna, Italy, July 2025.
13. S. Reddy, N. Choudhury, A. Hazarika and **T. Ojha**, "DyHSARW: A Dynamic GTS Scheduling Mechanism for Large IEEE 802.15.4 DSME-Based IoT Networks", in *Proceedings of IEEE WCNC*, Milan, Italy, March 2025.
14. S. Anand, N. Choudhury, **T. Ojha**, A. Hazarika, J. Dave, "Improving Network Efficiency in Clustered Tree Topology through PSO Optimization in IEEE 802.15.4-DSME based IoT Networks", in *Proceedings of IEEE ANTS*, Guwahati, India, December 2024.
15. C. Amala, B. Subbarao, **T. Ojha**, B. B. Das, S. K. Ram, and S. P. Mohanty, "An Off-chip Based PUF for Robust Security in FPGA Based IoT Systems", in *IEEE Proceeding on 22nd OITS International Conference on Information Technology*, Guntur, India, Dated: 12-14 December, 2024. (**Best Paper Award**)
16. **T. Ojha**, T. P. Raptis, M. Conti, A. Passarella, "Heterogeneity-aware P2P Wireless Energy Transfer for Balanced Energy Distribution", in *Proceedings of IEEE GLOBECOM*, pp. 4123-4128, Rio de Janeiro, Brazil, 2022.
17. **T. Ojha**, T. P. Raptis, M. Conti, A. Passarella, "Wireless Crowd Charging with Battery Aging Mitigation", in *Proceedings of IEEE SmartComp*, pp. 142 - 149, Helsinki, Finland, 2022.
18. **T. Ojha**, T. P. Raptis, M. Conti, A. Passarella, "MobiWEB: Mobility-Aware Energy Balancing for P2P Wireless Power Transfer", in *Proceedings of IEEE ISCC*, pp. 1 - 6, Athens, Greece, 2021. (Among **best papers** & Invited for extended version)
19. **T. Ojha**, S. Misra, N. S. Raghuwanshi, M. S. Obaidat, "iDVSP: Intelligent Dynamic Virtual Sensor Provisioning in Sensor-Cloud Infrastructure", in *Proceedings of IEEE GLOBECOM*, pp. 1 - 6, Abu Dhabi, UAE, 2018.
20. S. Bera, **T. Ojha**, S. Misra, M. S. Obaidat, "Cloud-based Optimal Energy Forecasting for Enabling Green Smart Grid Communication", in *Proceedings of IEEE GLOBECOM*, pp. 1 - 6, San Diego, CA, USA, 2015.
21. **T. Ojha**, S. Bera, S. Misra and N. S. Raghuwanshi, "Dynamic Duty Scheduling for Green Sensor-Cloud Applications," in *Proceedings of 6th IEEE International Conference on Cloud Computing Technologies and Science(CloudCom) Workshops*, pp. 841 - 846, Singapore, Dec. 2014.
22. **T. Ojha** and S. Misra, "MobiL: A 3-Dimensional Localization Scheme for Mobile Underwater Sensor Networks," in *Proceedings of 19th National Conference on Communications (NCC)*, pp. 1 - 5, New Delhi, India, IEEE, Feb. 2013.
23. **T. Ojha** and S. Misra, "HASL: High-Speed AUV-Based Silent Localization for Underwater Sensor Networks," in *Proceedings of the 9th International Conference on Heterogeneous Networking for Quality, Reliability, Security and Robustness (QShine)*, LNCS 115, pp. 128 - 140, Greater Noida, India, Springer, Jan. 2013.

DETAILS OF START-UP

Name: SkinCurate Research Pvt. Ltd.

Role: Co-founder and Director

Associated Duration: 2014-2021

Location: Kharagpur, India

Other Co-founders: Dr. Debdoot Sheet, Dr. Kausik Basak, Dr. Sri Phani Krishna Karri

Grants Received:

- BIRAC, DBT, Govt. of India (BIG grant): 41.79 Lakhs INR (Aug. 2015 - Jan. 2017)
- SINE, IIT Bombay and Intel Inc. (Plugin 2 Startup Cohort): 10.00 Lakhs INR (Oct. 2018 - Jun 2019)

Employment Generated: 4 (technical), 3 (non-technical)

Awards/Recognitions:

- 2018: 1st Runner-Up, CII Healthcare Innovation Summit
- 2018: Among Top 10 teams, Western Digital Innovation Bootcamp (along with Start-up India)
- 2018: Nominated for Economic Times Start-up Awards
- 2019: Indian Patent Published
- 2014: Selected for CAMTech Final round

PRODUCTS DEVELOPED

- “A mobile app-powered portable multi-spectral imaging prototype for skin disease detection”, at Indian Institute of Technology Kharagpur and SkinCurate Research Pvt. Ltd., 2013-16.
- “Jaltarang: A NS-3 based Underwater Sensor Network Simulator”, at Indian Institute of Technology Kharagpur, 2011-13. [URL: <https://cse.iitkgp.ac.in/smisra/swan/tre/doc/Jaltarang.zip>]
- “MAcoSim: Matlab-based Acoustic Underwater Simulator”, at Indian Institute of Technology Kharagpur, 2011-13. [URL: <https://cse.iitkgp.ac.in/smisra/swan/tre/doc/MAcoSim.zip>]

INVITED TALKS (SELECTED)

- January 2026: Invited talk on “AI/ML for the Next Evolution in Future IoT Networks” at 3rd Frontier Symposium in Data Science 2026 (FS-DSC 2026) at IISER Thiruvananthapuram.
- January 2026: Invited talk on “Evolution of IoT and Edge Computing” at FDP organized by SRM University-AP.
- November 2025: *Invited Talk* on “The Next Evolution in Future IoT Networks” at EPCI 2025, international conference organized by NIT Delhi and EICT, IIT Roorkee.
- September 2025: *Invited Talk* on “The Next Evolution in Future IoT Networks” at ACM ROCS organized by Chitkara University, Punjab.
- September 2025: *Invited Talk* on “Federated Learning Platforms Addressing Security and Privacy” at FDP organized by NIT Delhi and EICT, IIT Roorkee.
- July 2025: *Invited talk* on “On the Edge of Future IoT Networks: Sensing, Communication and Computation” at FDP organized by NIT Delhi and EICT, NIT Warangal.
- June 2025: *Invited talks* on “Sensing, Actuation, and Microcontroller/SBC” and “Machine Learning on Edge” at Workshop organized by iHUB Data, IIIT Hyderabad.
- July 2025: *Expert lectures* on “Energy Aspects of Future IoT Networks” and “Communication and Computing in Future IoT networks” at STP organized by MMTTC, IIT Patna at BIT Mesra.
- June 2025: *Invited talk* on “Future IoT Networks – Communication, Computation and Energy” at FDP organized by IIIT Kottayam and EICT, NIT Warangal.
- June 2025: *Expert lecture* on “Wirelessly Powered Future IoT Networks” and “Resource Management in 6G IoT networks” at STP organized by Malaviya Mission Teacher Training Centre (MMTTC), IIT Patna.
- May 2025: *Invited talk* on “Resource Management in Future IoT Networks” organized by IEEE CS Student Branch Chapter IIT Kharagpur and IEEE India Council.
- May 2025: *Invited talks* on “Energy aspects of Future IoT Networks” and “Communication and Computation in Future IoT Networks” at FDP organized by EICT, IIT Roorkee and Brainware University, Kolkata.
- April 2025: *Invited talks* on “Rise of the Edge Devices: Distributed Computing and Communication for 6G Networks” and “Are we really Empowered? Wireless-powered IoT for Futuristic Networks” at FDP organized by EICT, IIITDM Jabalpur and SRM University-AP.
- April 2025: *Expert lecture* on “Wirelessly Powered IoT Networks” and “Resource Management in 6G IoT networks” at STP organized by MMTTC, IIT Patna at K. K. Group of Institutes, Dhanbad.
- November 2024: *Guest lecture* on “Recent Trends and Future Directions in Internet of Things” at the IoT Course (UG), Dept. of EEE, BITS Pilani, Hyderabad campus.
- December 2023: *Invited talk* on “IoT and Edge Computing for Smart Agriculture” at webinar organized by Centurion University of Technology and Management, Odisha.
- December 2023: *Resource Person* at ATAL FDP on “Integrated IoT and Machine Learning Methods for Smart Sustainable Cities” organized by Vignan’s University, Guntur.
- June 2023: *Resource Person* at FDP on “Applications of Machine Learning and the Internet of Things in Smart Cities” organized by EICT, NIT Warangal and Techno College of Engineering, Agartala.
- Dec. 2022: *Keynote Speaker* at ICACIE 2022 conference, Cuttack, India.

PROFESSIONAL MEMBERSHIPS

- *IEEE* [ComSoc, CompSoc]: Senior member (2023–present), Member (2021–2023), Student (2013–2020)
- *ACM*: Professional member (2021–present), Student member (2013–2020)

ADMINISTRATIVE RESPONSIBILITY

- *Faculty In-Charge* for ‘Department Labs’ at Dept. of Maths & Computing, IIT (ISM) Dhanbad (August 2025–).
- *Faculty In-Charge* for ‘Branding of Departments’ at IIT (ISM) Dhanbad (May 2025–).
- *Coordinator* for department website maintenance at IIT (ISM) Dhanbad (February 2025–).
- *Editor* for the department newsletter, GANANAM, at IIT (ISM) Dhanbad (October 2024–).

VOLUNTEERING EXPERIENCE

- *Member*, IEEE IC Technical Activities (2026–)
- *Mentor*, ACM Anveshan Setu Fellowship program (2024–)
- *Member*, ACM India Research Facilitation Committee (RFC) (2024–)
- *Member*, IEEE India Council Awards Committee (2024)
- *Student Representative*, IEEE Computer Society India Council SAC (2017)
- *IEEE UPP Liaison*, IEEE Student Branch, IIT Kharagpur (2016–2017)
- *Chair* of Executive Committee, IEEE Student Branch, IIT Kharagpur (2015–2016)

REFeree SERVICE

- *Journal Editorial Board*:
 - IEEE Transactions on Green Communications and Networking
 - IEEE Networking Letters
 - Scientific Reports (Springer Nature)
- *Examiner / Reviewer*:
 - IEEE India Council Awards, 2023; 2024
- *Conference Organizing Committee member*:
 - *Chair*, 6DCIoT 2026 workshop (with IEEE PerCom 2027), Goa, India.
 - *Chair*, Travel and Visa, IEEE ANTS 2026.
 - *Chair*, 6DCIoT 2025 workshop (with IEEE ANTS 2026), IIT Roorkee, India.
 - *Chair*, 6DCIoT 2026 workshop (with IEEE VTC Spring 2026), Nice, France.
 - *Chair*, 6DCIoT 2025 workshop (with IEEE ANTS 2025), New Delhi, India.
 - *Co-Chair*, Metaverse Futures workshop (with IEEE FNWF 2025), Bangalore, India.
 - *Chair*, 6DCIoT 2024 workshop (with IEEE ANTS 2024), Guwahati, Assam.
 - *Publicity Co-Chair*, IEEE DCOSS 2022, California, USA.
 - *General Chair*, WPSN 2021 (in conjunction with IEEE DCOSS), Pafos, Cyprus.
 - *Publicity Co-Chair*, IEEE DCOSS 2021, Pafos, Cyprus.
- *Technical Program Committee member* (Selected only):
 - *IEEE PIMRC*: 2018–2026
 - *IEEE GLOBECOM*: 2025, 2023
 - *IEEE VTC*: Spring 2025
 - *IEEE ISCC*: 2021–24

- *Journal reviewer*:
 - IEEE Transactions (TNSM, TMC, TSC, TVT)
 - IEEE Journals/Letters (IoT J., Systems J., Sensors J., Embedded System L., Access)
 - ACM Transactions (TOSN)
 - Elsevier (Ad Hoc Net., ComNet, ComCom, PMC, CompAg)
- *Conference reviewer* (Selected only):
 - IEEE PIMRC (2017–2025); IEEE ISCC (2021–2024); IEEE GLOBECOM (2023–25); IEEE CCNC (2021, 2023)
 - Others: IEEE LCN, IEEE DCSS, IEEE ATC, COVI-COM (IEEE ICC), IEEE ICCCS, IEEE ANTS, IEEE NCC

LANGUAGES

ENGLISH:	Fluent	BENGALI:	Mothertongue
HINDI:	Fluent	ITALIAN:	Basic Knowledge (A1)

REFERENCE

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